

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

PARADIGM JUSTIFICATION STATEMENT

DELIVERABLE No. 1

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Table of Contents

1. Research Topic, Scope, and Context	3
2. Refined Research Questions.....	3
3. Philosophical Position and Research Approach	4
4. Why Design Science Research	5
5. Participatory Co-Design and Ethics	5
6. Operationalization of Functional Communication	6
7. Evaluation Rigor and Comparator	6
8. Implications of the Paradigm Choice	6
9. One Remaining Tension	7
References	8

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DELIVERABLE No. 1

Research topic: A Human-in-the-Loop Multimodal AAC Prototype in Spanish to Support Functional Communication in Autistic Children Requiring Very Substantial Support

1. Research Topic, Scope, and Context

Many autistic children who are minimally speaking or nonspeaking and require very substantial support cannot reliably express needs, preferences, discomfort, emotions, or requests for help through speech alone. In Peru, this challenge is intensified by the limited availability of Spanish-language augmentative and alternative communication (AAC) tools that adapt to a child's routines, family vocabulary, and sociocultural context. The project therefore focuses on a deliberately scoped Design Science Research study: the design and pilot validation of a functional human-in-the-loop multimodal AAC prototype in Spanish. To keep the work feasible within three years, the thesis will focus on one clearly defined user subgroup, one structured therapeutic communication context, and two to three primary communication outcomes rather than full-scale clinical validation across home, school, and therapy settings.

2. Refined Research Questions

- Design question: How can a human-in-the-loop multimodal AAC prototype in Spanish be designed to provide contextually relevant communication options while preserving user agency and personalization?
- Evaluation question: To what extent does the prototype improve independent functional communication, communicative diversity, and the level of assistance required when compared with a predefined static AAC condition?
- Risk and ethics question: What privacy, autonomy, bias, homogenization, and safety risks emerge during use, and what safeguards are required to manage them responsibly?

3. Philosophical Position and Research Approach

My philosophical position is Pragmatism. The study is anchored in pragmatism: knowledge is judged by its capacity to explain and improve action in a real problem context. This position is compatible with DSR because the study does not seek knowledge only by observing the world; it also creates and evaluates an artifact intended to improve a consequential human activity.

Table 1

Philosophical and Methodological Positioning of the Study

Dimension	Position in this study
Ontology	Communication barriers and system effects are real, but their meaning, usefulness, and acceptability are context-dependent and experienced differently by children, caregivers, and professionals.
Epistemology	Valid knowledge comes from converging evidence: observable system and communication outcomes, rigorous artifact evaluation, and situated stakeholder judgments about usefulness, safety, and contextual relevance.
Methodology	Design Science Research provides the governing build–evaluate–iterate logic. Quantitative repeated-measure evidence and focused qualitative stakeholder feedback are used inside the evaluation cycle, not as a second governing methodology.
Axiology	The study prioritizes dignity, autonomy, privacy, non-maleficence, fairness, communicative diversity, and the child’s right to reject, ignore, or stop interaction with the system.

Note. Author’s own elaboration based on Guba and Lincoln (1994) for the philosophical positioning and Hevner et al. (2004) for the Design Science Research framework.

4. Why Design Science Research

Design Science Research (DSR) is the primary research approach because the central doctoral contribution is the design, development, rigorous evaluation, and refinement of a novel multimodal AAC artifact, together with transferable design knowledge for safe contextual personalization. The study follows a build–evaluate–iterate logic and a human-in-the-loop architecture.

A purely positivist approach can estimate effects but cannot, by itself, account for how the artifact should be designed and refined.

A purely interpretivist approach can illuminate lived experience and context but cannot establish technical performance or repeated functional communication outcomes. Mixed evidence remains important, but it serves the evaluation and refinement of the artifact rather than replacing DSR as the governing research logic. The intended contribution is therefore not merely “an app that works,” but an evaluated artifact plus design principles that can inform future responsible AAC systems.

5. Participatory Co-Design and Ethics

The design process will incorporate participatory input from caregivers, speech-language professionals, and autistic AAC users where appropriate, including autistic adults who can contribute lived-experience perspectives without placing unnecessary burden on child participants. Before any user study, the project will obtain institutional ethics approval, written caregiver consent, and child assent using developmentally and communication-appropriate procedures.

For minimally speaking or nonspeaking participants, assent will include observable willingness to engage and predefined withdrawal or distress signals; participation will stop when those signals appear. Data collection will follow minimization principles, collect only information required for the research questions, separate identifiers from research records, restrict access, and define retention/deletion rules in advance. On-device processing will be preferred whenever technically feasible, especially for photographs, voice-related data, and interaction logs. The system will not infer thoughts or communicate on behalf of the child; it will only propose options that can be selected, rejected, or ignored.

6. Operationalization of Functional Communication

For this study, functional communication is defined as an intentional communicative act that conveys a meaningful request, rejection, choice, comment, request for help, discomfort, or affective state to an interlocutor within the target activity. The coding scheme will explicitly distinguish independent/spontaneous acts from prompted acts.

The three primary outcomes will be: (1) independent functional communication acts per session; (2) diversity of communicative functions or concepts expressed; and (3) level of assistance or prompting required. Message-selection time will be treated as a secondary descriptive measure, not as an unquestioned success criterion, because faster interaction is not necessarily better when the objective is to preserve the child's own communicative agency.

7. Evaluation Rigor and Comparator

The initial evaluation plan is a small single-case pilot with a target minimum of four participants and repeated observations in one structured therapeutic communication context. Because the thesis includes a direct comparison with static AAC, an alternating-treatments single-case design is the leading option: the intelligent prototype and a predefined static AAC comparator would be presented in counterbalanced or randomized order where feasible, with repeated measurement across sessions.

The protocol will prespecify the static comparator, session procedures, prompting hierarchy, fidelity checklist, safety/stopping rules, and analysis plan. A prespecified subset of sessions will be independently double-coded to estimate inter-observer agreement, and treatment fidelity will be documented. Individual trajectories will be reported before aggregate summaries so that meaningful gains or harms are not hidden by group averages.

8. Implications of the Paradigm Choice

- Data: interaction logs, coded functional communication acts, communicative diversity, assistance/prompt level, rejected or modified AI suggestions, and focused caregiver/specialist feedback.
- Artifact: a Spanish-language multimodal AAC prototype integrating photographs/pictograms, text, speech output, contextual information, and human validation.

- Contribution: a rigorously evaluated artifact plus reusable design principles for responsible, context-aware, human-in-the-loop personalization.
- Target venues: ACM ASSETS, ACM Transactions on Accessible Computing, and Augmentative and Alternative Communication.
- Localization: established open pictogram resources and Peruvian Spanish speech synthesis will be assessed early to reduce avoidable technical and cultural risk.

9. One Remaining Tension

The central tension is how to produce credible evidence of benefit with a small, heterogeneous, and vulnerable pediatric population without overstating generalizability. The response is to narrow the claim: this thesis will seek rigorous pilot evidence about whether the artifact is useful, safe, and promising under explicitly bounded conditions. It will not claim broad clinical effectiveness. Repeated within-participant evidence, transparent reporting of individual trajectories, fidelity and inter-observer checks, and stakeholder interpretation will be used to support appropriately limited conclusions.

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